

LLM4Rec Homework Report — Assignment A03

Collaborative Filtering Under Sparse Evidence: Comparing User-Based and Item-Based Movie Recommendation

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Assignment: A03 — Collaborative Filtering (Week 3)

Abstract

This assignment implements User-Based and Item-Based collaborative filtering on MovieLens 100K using co-rated-only cosine similarity. I first audited and verified the starter implementation, then compared the two methods under leave-one-out evaluation and analyzed how sparse co-rating support affects similarity reliability. User-Based CF achieved 0.988% HitRate@5 while Item-Based CF achieved 0%; the median overlap was 10 ratings for user pairs but only 1 for item pairs. An overlap-weighting experiment improved evidence support but not recommendation quality consistently. The main lesson is that a high similarity score is not necessarily a reliable one when it is estimated from very little shared evidence.

Index Terms — collaborative filtering, cosine similarity, MovieLens 100K, sparsity, User-Based CF, Item-Based CF

1. Introduction

The Week 3 task is to implement User-Based and Item-Based collaborative filtering on the same MovieLens 100K data used in Week 2, choose a consistent missing-value strategy, and compare recommendation quality, efficiency, and behavior under sparse ratings. Unlike Week 2, which uses movie genres, collaborative filtering relies only on collective rating patterns.

During testing, I found that high cosine values were often supported by very few co-ratings. For example, MovieLens user 2 receives a User-Based prediction of 5.000 for *The Godfather: Part II* (1974), but only two of the top-20 neighbors rated that movie. I therefore evaluated not only prediction scores but also the amount of evidence behind them.

The report contributes:

- a verified User-Based and Item-Based CF implementation with side-by-side rating prediction and Top-5 recommendation;
- reliability, quality, and efficiency diagnostics for sparse user–item interactions;
- a controlled overlap-weighting experiment and an aligned comparison with the Week 2 content-based recommender.

Repository: <https://github.com/chenfengru/LLM4Rec-Course>.

2. Related Work

Collaborative filtering estimates preferences from user–item interaction patterns rather than item metadata. Resnick et al. introduced an early user-neighborhood formulation [1], while Sarwar et al.

developed Item-Based CF using similarities between rating patterns [2]. Neighborhood methods are simple and interpretable but sensitive to sparsity and insufficient overlap [3]. MovieLens 100K provides the explicit user–movie ratings used here [4].

I also considered mean imputation, overlap-weighted similarity, and matrix factorization. Co-rated-only cosine was retained as the required transparent baseline; overlap weighting was tested separately as a controlled improvement, while the other alternatives are discussed conceptually rather than reported as empirical results.

3. Method

The workflow was:

starter audit → rating matrix → User/Item CF → reliability tests → overlap-weighting experiment.

3.1 Co-Rated Cosine Similarity

MovieLens contains 943 users, 1,682 movies, and 100,000 ratings. The matrix uses $R_{ui} = 0$ only to represent an unobserved interaction; actual ratings are 1–5.

The chosen missing-value strategy is **co-rated entries only**. For vectors a, b ,

$$C(a, b) = \{k : a_k \neq 0, b_k \neq 0\},$$

$$s(a, b) = \frac{\sum_{k \in C} a_k b_k}{\sqrt{\sum_{k \in C} a_k^2} \sqrt{\sum_{k \in C} b_k^2}}. \quad (1)$$

No-overlap or zero-denominator cases return 0. Missing ratings are not imputed.

3.2 User-Based and Item-Based Prediction

For User-Based CF, the 20 most similar users with positive similarity form the neighborhood. An unseen movie i is predicted as

$$\hat{r}_{ui}^U = \frac{\sum_{v \in N(u): R_{vi} > 0} s(u, v) R_{vi}}{\sum_{v \in N(u): R_{vi} > 0} s(u, v)}. \quad (2)$$

For Item-Based CF, candidate movie i is compared with movies already rated by u :

$$\hat{r}_{ui}^I = \frac{\sum_{j: R_{uj} > 0, s(i, j) > 0} s(i, j) R_{uj}}{\sum_{j: R_{uj} > 0, s(i, j) > 0} s(i, j)}. \quad (3)$$

Already-rated movies are excluded and the five highest-scoring unseen movies are returned.

3.3 Evaluation and Improvement

For leave-one-out evaluation, each eligible user’s most recent rating ≥ 4 is hidden before prediction. The main protocol includes 911 users and reports HitRate@5 and MRR@5.

To test whether small overlap explains unstable similarities, I also evaluated

$$s_T(a, b) = s(a, b) \min \left(\frac{n_{ab}}{T}, 1 \right), \quad T \in \{5, 10, 20\}, \quad (4)$$

where n_{ab} is the number of co-rated entries. This is an experimental confidence adjustment; the final application keeps raw co-rated cosine.

4. Experiments

4.1 Verification

The final application reproduced 943 users, 1,682 movies, and 100,000 non-zero ratings. Small deterministic matrices were used to verify cosine, no-overlap, one-overlap, User-Based prediction, and Item-Based prediction. Regression checks confirmed that Top-5 lists never include already-rated movies and contain only finite scores.

The starter audit also found the same semantic parser defect seen previously: 19 MovieLens genre flags were mapped to only 18 labels because `unknown` was missing. After repair, Toy Story maps exactly to Animation/Children's/Comedy and Western-tagged movies remain reachable. The defect does not affect the CF calculations themselves because they use ratings rather than genres.

4.2 Reliability, Quality, and Efficiency

Table 1: Co-rating support in MovieLens 100K.

Pair type	Unique pairs	Median overlap	≤ 2
User-user	444,153	10	14.29%
Item-item	1,413,721	1	60.38%

Among similarities with cosine at least 0.9, 11.94% of user pairs and 47.70% of item pairs still had at most two co-ratings. A cosine of 1 can therefore be mathematically valid even when it is supported by only one shared rating.

Table 2: Leave-one-out evaluation and measured runtime.

Method	HitRate@5	MRR@5	Mean runtime
User-Based CF	0.988%	0.004848	1.59 ms
Item-Based CF	0.000%	0.000000	388.48 ms

The runtime comparison is specific to this JavaScript implementation. Here, the item space is larger than the user space: there are 1,413,721 unique item-item pairs versus 444,153 user-user pairs, so the usual “item-based is cheaper because there are fewer items” intuition does not hold for MovieLens 100K.

4.3 Improvement and Week 2 Comparison

Overlap weighting increased the median evidence behind contributing User-Based similarities from 1 to 14 at $T = 10$, but Top-5 quality was not stable:

	Raw	$T = 5$	$T = 10$	$T = 20$
User-Based hits	9	11	10	5
Item-Based hits	0	0	0	0

I therefore kept raw cosine as the final method rather than selecting a threshold from the evaluation cohort.

For 853 users aligned with Week 2, the content-based three-movie profile achieved 2.34% HitRate@5, User-Based CF 1.41%, and Item-Based CF 0%. Their Top-5 sets were also almost

disjoint: Content/User-Based shared only 0.081 movies out of five on average. This reflects the different signals: Week 2 ranks genre similarity, whereas Week 3 ranks rating behavior.

5. Discussion

5.1 BUILD

I first ran and audited the starter code, then implemented raw co-rated cosine for both User-Based and Item-Based CF. Deterministic examples verified the similarity and prediction equations before large-scale evaluation.

5.2 ASK WHY

The first real-user results contained many 5.000 predictions and very high item similarities. Instead of interpreting these scores as strong evidence, I asked how many shared ratings actually supported them.

5.3 TEST

I measured overlap for every user and item pair, reconstructed prediction scores from their contributors, ran the 911-user leave-one-out evaluation, and independently checked selected held-out ranks and final UI outputs.

5.4 EXPLAIN

The main issue is sparsity rather than an incorrect cosine implementation. With positive 1–5 ratings, one common observation can already produce cosine 1. Item pairs are especially fragile because their median overlap is only 1.

The starter parser defect had a separate root cause: the first MovieLens genre flag, `unknown`, was missing from the label array. Adding it restored the correct one-to-one mapping. This was verified using known movie genres.

For missing values, co-rated-only similarity is simple and uses only observed data, but it can be unreliable under small overlap. Mean imputation introduces artificial averages; overlap weighting adds an explicit confidence assumption; matrix factorization is more expressive but also more complex and expensive. A completely new user or item remains a cold-start case because no interaction evidence exists.

5.5 IMPROVE

The controlled improvement multiplied cosine by an overlap-confidence factor. It substantially improved the evidence supporting User-Based similarities but did not produce a stable HitRate improvement and did not fix Item-Based CF. The experiment therefore clarified the failure mode without justifying a change to the final baseline.

With more time, I would repeat the comparison using a global chronological split and test a confidence-aware or mean-centered neighborhood method without tuning on the evaluation set.

6. AI Usage Disclosure

AI tools used. The formal coding workflow used OpenAI Codex (CLI 0.157.1). ChatGPT (GPT-5.6 Sol) was used separately for planning experiments, reviewing evidence, checking assignment coverage, and drafting the report.

How AI was used. AI assisted with repository inspection, implementation scaffolding, debugging, analysis scripts, regression tests, UI iteration, and report drafting.

What I personally verified. I checked the dataset dimensions and byte-identical Week 2/3 files, verified the Toy Story genre repair, checked cosine and prediction formulas on small examples, reproduced already-rated-item exclusion and Top-5 outputs, and reran the final regression verification against the stored results.

What I trusted without full independent verification. I did not independently audit every non-critical line of code or every AI tool call. Mean imputation and matrix factorization were discussed conceptually rather than implemented, and the reported quality results were not tested under a global chronological split.

Session notes. The submitted `session.json` is the unmodified native Codex audit trail for the formal Week 3 workflow.

References

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